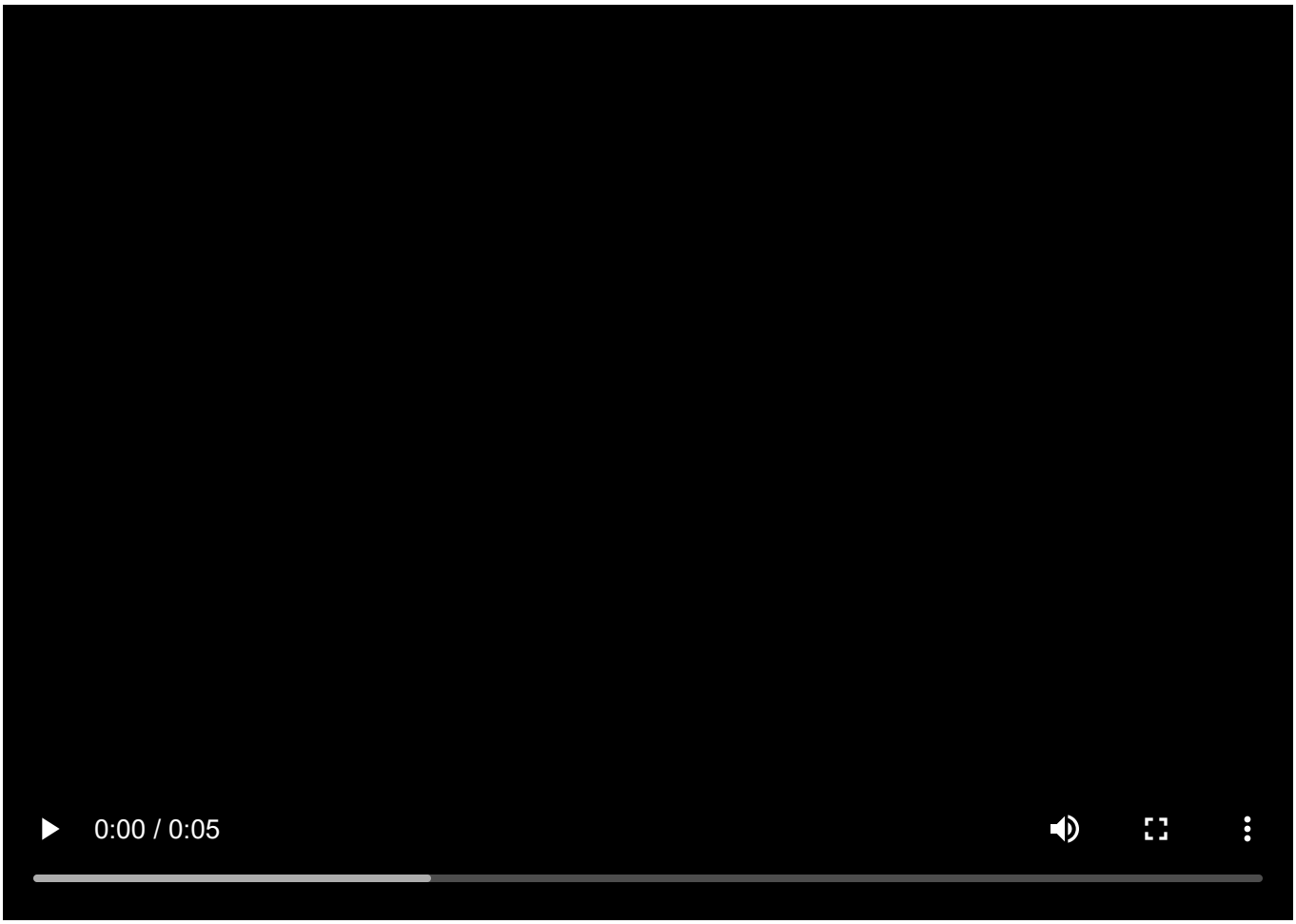
Starting Point - Used Lab 09 FSM Single Stoplight Simulation as a Template

(1) Use the FSM as a central control for both traffic lights.
This implemented the logic that the three states from traffic light A could be used to control states in traffic light B.

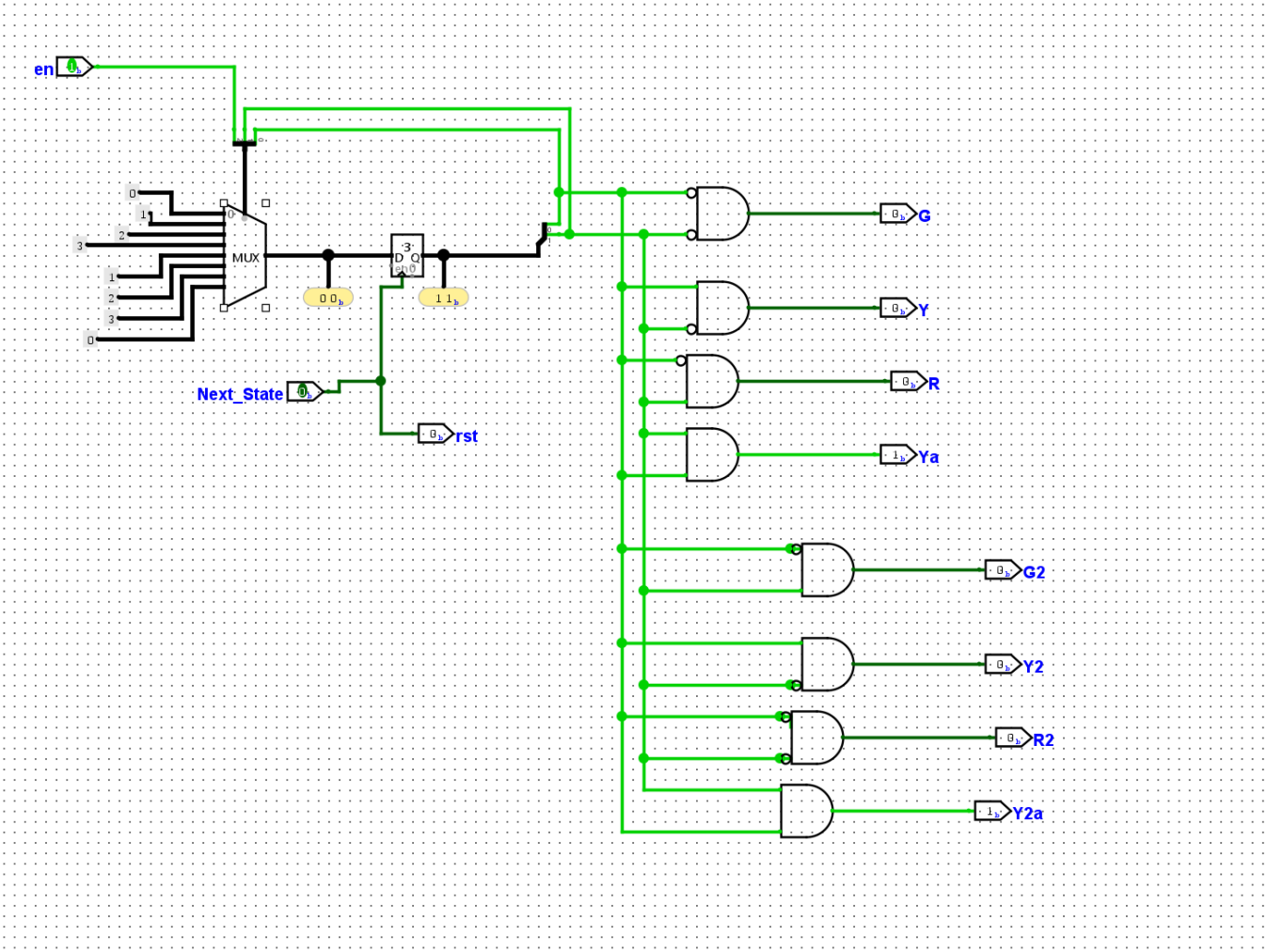
- Red_A and Green_B - 00
- Yellow_A and Yellow_B - 01
- Green_A and Red_B - 10

This logic preliminarily works but causes a yellow state to be skipped when going from 10 to 00.




0:00 / 0:05

Added a fourth state to the lights to create a yellow stage in between Light 2 and Light 1.



The FSM switches between the 4 states at the rate of the clock.
The MUX if EN=0 the inputs point to holding the same state, if EN=1 the inputs point to the next state.



Behavior Fix - Stopights only go to yellow when moving from Green to Red

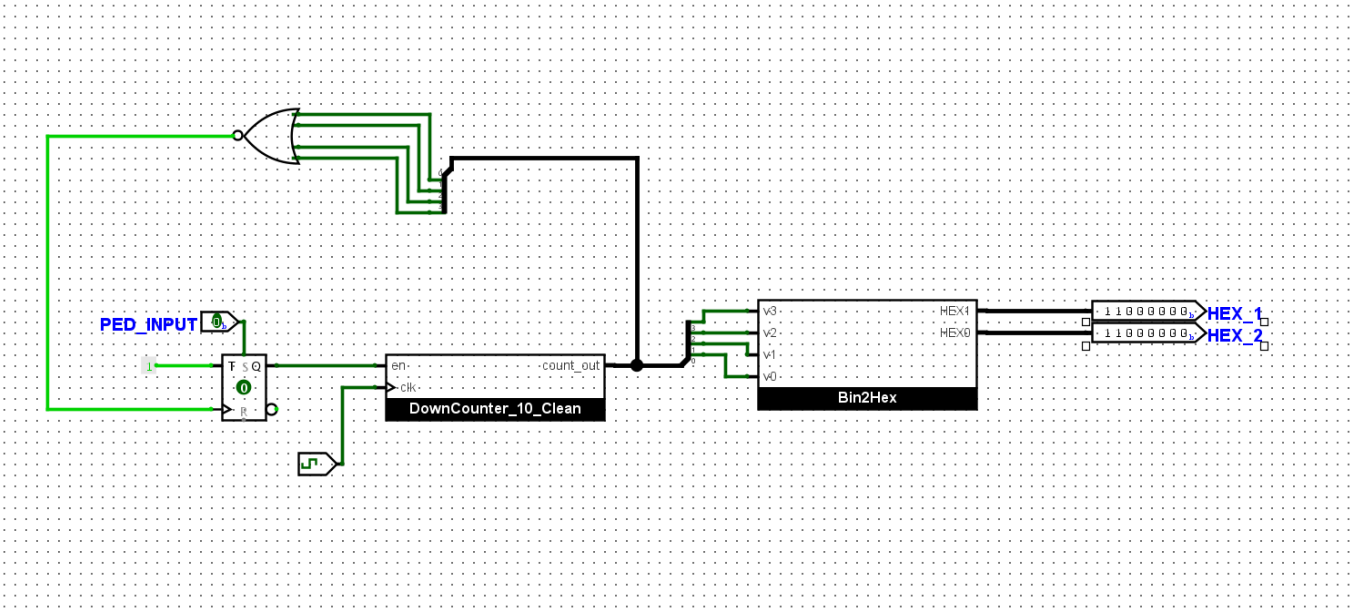


A light holds RED when the other light is both GREEN and YELLOW.



(2) IMPLEMENATION - HOLD GREEN UNTIL PED BUTTON IS PRESSED

Creation of PED Button to trigger a 10 second count down



Pedestrian buttons trigger a 10 second count down

(3) EN is triggered when PED Counter = 00



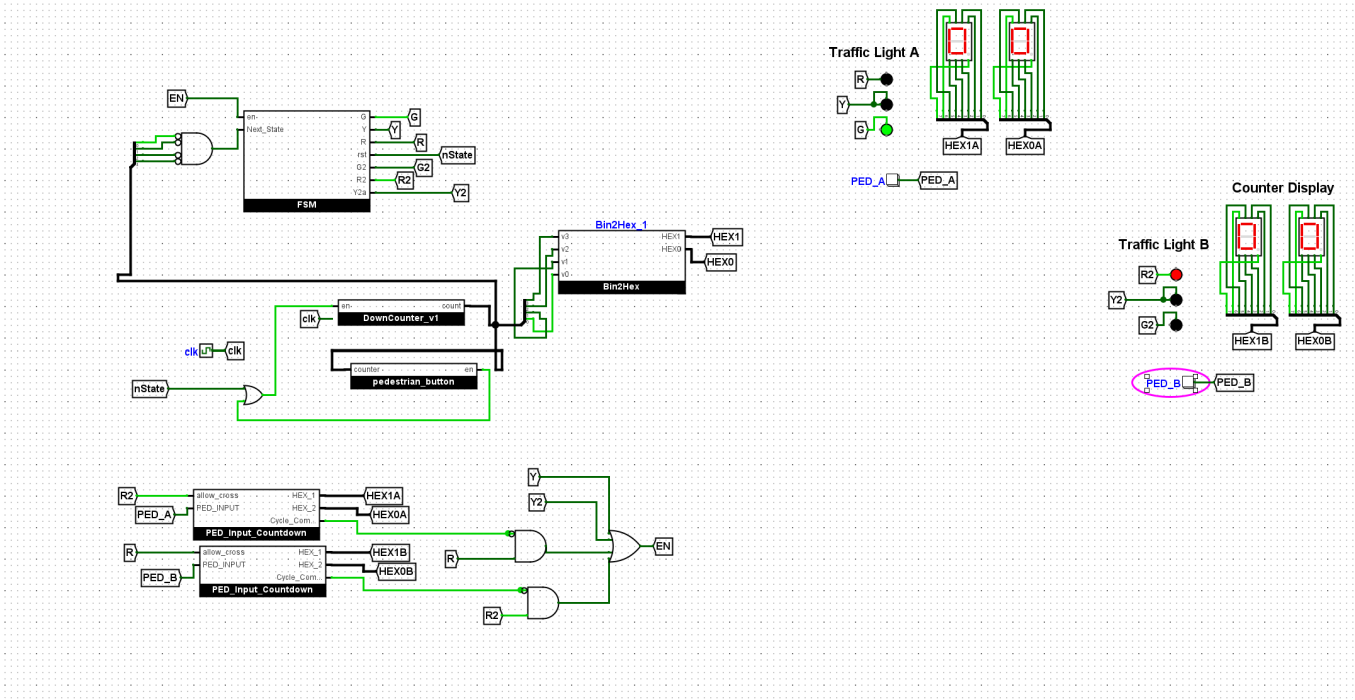
Currently Cycle_Complete gives 1 when counter reaches zero. When both Cycle_Counters are at 0 the stoplghts continue through their automatic cycle.

**(4) When a traffic light reaches RED it holds until the opposite PED button is pressed



EN Pin is only 1 if

- Stoplights are yellow -> move to red state
- Cycle complete = 1 AND the opposite light is Red
  - ISSUE - If a pedestrians press both buttons the system doesn't know how to "wait" for the other cycle to conclude
  - GOOD - If a pedestrian presses the same button twice the light holds its position so the pedestrian can cross
  - ISSUE - The timing of the cycle is off -> the pedestrian should only cross when the oposite light is RED not yellow.



Created an "ALLOW" crossing function that should only be triggered when the opposite light from the PED signal is pressed. However, currently when PED B is pressed no countdown is triggered

Possible Issues

- Timing mechanism is still tied to a three second transition between states and there is unnecessary hardware.



Counter only starts once the Green for the same crossing direction is achieved

GOAL ACHIEVED

Limitation - Cycle that traffic light goes through is FAST, however it does signal a traditional traffic cycle prior to the button being pressed then prioritizes the pedestrian crossing.